POSITIVITY OF LOCAL INTERSECTION MULTIPLICITIES ON NONSINGULAR VARIETIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B0U: sources/stacks-chow/tags.txt, source lines 13872–13872. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B0Q: sources/stacks-chow/tags.txt, source lines 13868–13868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B04: sources/stacks-chow/tags.txt, source lines 13847–13847. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let X be a nonsingular variety. Let $V \subset X$ and $W \subset Y$ be closed subvarieties which intersect properly. Let $Z \subset V \cap W$ be an irreducible component. Then $e(X, V \cdot W, Z) > 0$.*

Proof. By Lemma 0B0U we have

$$e(X, V \cdot W, Z) = e(X \times X, \Delta \cdot V \times W, \Delta(Z))$$

Since $\Delta : X \to X \times X$ is a regular immersion (see Lemma 0B0Q), we see that $e(X \times X, \Delta \cdot V \times W, \Delta(Z))$ is a positive integer by Lemma 0B04. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.